

EGOR SEREBRIAKOV

APPLIED MACHINE LEARNING · DATA SCIENCE

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Summary

M.Sc. student in Artificial Intelligence and researcher at Mila with a strong background in **data science, machine learning, optimization, and automation**. Trained through scientific research in Physics and AI, I combine a rigorous analytical mindset and a practical approach to real-world problems. I have hands-on experience in:

- (i) **Data analysis and validation:** applying Python-based analysis to assess data quality, consistency, and reliability.
- (ii) **End-to-end data workflows:** building and maintaining data-driven solutions, from requirements definition and data preparation to validation, deployment, and ongoing monitoring.
- (iii) **Applied modern machine learning:** from classical ML and deep models for prediction, classification, and anomaly detection to large language models and agentic workflows, with a focus on validation, robustness, and efficiency.

Skills

Programming	Python (<i>PyTorch, NumPy, pandas, scikit-learn, OOP</i>), C
Data Analysis	Exploratory Data Analysis (EDA), Data Validation, Statistical Analysis, Feature Engineering
Machine Learning	LLMs, Agentic AI, Supervised & Unsupervised Learning, Model Validation, Evaluation Metrics, Robustness, Optimization
Tools	Git, Linux, Bash, Jupyter, VS Code
Soft Skills	Communication, Teamwork, Adaptability, Responsibility, Emotional Intelligence

Education

ITMO University	<i>St. Petersburg, Russia (Remote)</i>
M.Sc. STUDENT IN ARTIFICIAL INTELLIGENCE, GPA 4.00	Sep 2025 - Present
- Research on optimization and robustness of large physics-informed neural networks. - Selected Coursework: <i>Mathematical Statistics, ML Systems Design, Deep Generative Models.</i>	
Brown University	<i>Providence, RI</i>
M.Sc. IN PHYSICS	Sep 2022 - May 2024
- Research on deep learning models for classification and anomaly detection in particle physics, in collaboration with CERN. - Selected Coursework: <i>Deep Learning, Applied Machine Learning and AI.</i>	
Moscow State University	<i>Moscow, Russia</i>
B.Sc. IN PHYSICS WITH HONORS, GPA 3.92	Sep 2018 - May 2022

Projects & Experience

Mila, Québec AI Institute – CERC-AAI Lab	<i>Montréal, QC</i>
RESEARCH COLLABORATOR – LLMs & Agentic Systems	Mar 2026 - Present
- Investigating LLM generalization mechanisms in time-series through gradient alignment analysis, examining how gradient structure across data distributions relates to transfer behavior. - Studying autonomous AI agents, including orchestration frameworks and AI-scientist architectures, with emphasis on tool use, self-evaluation, monitoring, and safe deployment of autonomous workflows.	
ITMO University – Optimization of Physics-Informed Neural Networks (PINNs)	<i>St. Petersburg, Russia (Remote)</i>
RESEARCH ASSISTANT – Data Analysis / Applied ML	Sep 2025 - Apr 2026
Developed reinforcement learning–based methods to automate and optimize training dynamics of PINNs in strongly nonlinear systems, improving convergence stability through adaptive hyperparameter control.	
Brown University – Transfer Learning for Jet Tagging in Particle Physics	<i>Providence, RI</i>
RESEARCH ASSISTANT – Data Analysis / Applied ML	Jan 2024 - Jun 2024
- Investigated knowledge distillation by transferring information from a large-scale graph neural network (GNN) teacher model to student models trained in two settings: with reduced input resolution and with simplified architecture. - Achieved >3x inference speedup through teacher-student distillation while preserving classification performance (AUC ≈ 0.9), enabling faster jet tagging with the ultimate goal of developing models suitable for real-time deployment at CERN.	

Brown University – Anomaly Detection at the Large Hadron Collider

Providence, RI

RESEARCH ASSISTANT – Data Analysis / Applied ML

Jan 2024 - Jun 2024

- Designed and implemented **unsupervised anomaly detection pipelines** using deep autoencoders for high-dimensional data.
- In controlled synthetic stress scenarios, detected **large anomalous regimes** (up to **~50% of samples**), and used these results to analyze model robustness, failure modes, and false-positive risk.

Professional Development

AI Talent Hackathon 2025

Online

BACKEND & ML DEVELOPER

Sep 2025

- Developed backend services and data-processing pipelines for an education platform, integrating a language model for automated grading.
- Designed a rubric-based feedback system that generates structured, personalized feedback for each user.
- Implemented a chat assistant and engagement mechanisms (points, badges), and evaluated their impact during live demonstrations.

Princeton University

Princeton, NJ

CERN COMPUTATIONAL & DATA SCIENCE SUMMER SCHOOL STUDENT

Jul 2024

- Completed hands-on training in parallel programming, optimization, and collaborative software development.
- Worked with CERN and Princeton University researchers on data science and machine learning techniques for high energy physics.

Brown University

Providence, RI

TEACHING ASSISTANT

Sep 2022 - May 2024

- Teaching Assistant for four physics courses: mentored students during lab sessions, graded coursework, and led problem-solving workshops.

Selected Awards

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| 2025 | Best EdTech Project , AI Talent Hackathon 2025 | Online |
| 2025 | Winner , Junior ML-Contest 2025 | St. Petersburg, Russia |
| 2022 | Diploma with Honors , Moscow State University (Top 7%) | Moscow, Russia |
| 2019 | Academic Excellence Scholarship , Moscow State University | Moscow, Russia |

Presentation

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| Jan 2026 | AI Talent Conf 2026 , Reinforcement Learning for Physics-Informed Neural Networks | Online |
| Jun 2025 | ML-Contest 2025 , Transfer Learning / Knowledge Distillation for Jet Tagging | St. Petersburg, Russia |
| May 2024 | Deep Learning Day @ Brown , Machine Learning for Particle Physics | Providence, RI |

Extracurricular Activity

MSU Academic Group No. 15

Moscow, Russia

ELECTED REPRESENTATIVE

Sep 2018 - May 2022

- Represented and protected the interests of a group of 24 students; upheld student rights, ensuring a respectful and inclusive school environment.
- Organized and coordinated student–faculty communication and logistics at the beginning of the pandemic.

Beloretsk Center of Extracurricular Activities

Beloretsk, Russia

VOLUNTEER TEACHER

Oct 2017 - Jun 2018

- Instructed high school students in mathematics and physics over two semesters.